Abhinav Mishra

• +91 8179401639 • myworkspace.mishra@gmail.com • LinkedIn • GitHub • Hyderabad, Telangana

Full Stack Python Developer with hands-on experience building backend services, REST APIs, AI-powered applications, automation workflows, and full-stack applications using Python, FastAPI, JavaScript, React, SQL, Firebase, and Streamlit. Experienced in developing secure APIs, interactive web interfaces, database integrations, authentication, document processing, third-party API integrations, and cloud-deployed applications. Proficient in LangChain, LangGraph, LLM APIs, RAG, n8n, Git, and modular software development with a strong foundation in Object-Oriented Programming, data structures, debugging, testing, and scalable application design.

PROFESSIONAL EXPERIENCE

Zeflo Solutions — Intern **(Nov 2025 – Jan 2026)**

- Built automation workflows using **n8n, REST APIs, AI services, databases, and third-party applications** to automate business processes and application tasks.
- Developed **customer feedback automation and AI-powered response generation** by integrating multiple services through API-based workflows.
- Integrated external APIs with **data validation, JSON processing, and exception handling** to improve application reliability and workflow execution.
- Debugged workflow failures and optimized automation logic while collaborating with cross-functional teams using **Git and structured testing practices**.

Marut Drone Tech Pvt. Ltd. — Intern **(Aug 2025 – Nov 2025)**

- Supported **software development, system testing, hardware validation, and debugging** for engineering applications.
- Performed **defect analysis and troubleshooting** to identify system issues and improve application and system reliability.
- Tested **software-hardware integrations** and analyzed system behavior to identify implementation and performance issues.
- Documented technical findings and collaborated with engineering teams to **implement and validate corrective solutions**.

TECHNICAL SKILLS

Programming:	Python, JavaScript, SQL, HTML5, CSS3, OOP, Data Structures, Algorithms
Software Development:	FastAPI, REST APIs, JSON, API Integration, Authentication, Exception Handling, File Handling, Modular Architecture
Frontend:	React, Responsive UI, Streamlit
AI / GenAI:	LangChain, LangGraph, Google Gemini API, Groq LLM, Claude API, RAG, Prompt Engineering, AI Agents
Databases:	Firebase Firestore, PostgreSQL, MySQL, SQLite
Automation:	n8n, Workflow Automation, REST API Automation, Third-Party API Integration
Libraries:	Pandas, OpenPyXL, PyPDF, Python-Docx, Pillow, BeautifulSoup
Development Tools:	Git, GitHub, Postman, VS Code, Docker
Testing / Engineering:	Debugging, Testing, Version Control, Agile Development, Software Engineering
Cloud / Deployment:	Streamlit Community Cloud, Environment Variables

PROJECT EXPERIENCE

Multi-Tool AI Agent ***[TOOL]***

- Developed a production-ready Full Stack AI application using Python, Streamlit, LangGraph, Groq LLM, Firebase, and REST APIs with intelligent tool routing.
- Built secure authentication, Firestore database integration, AI-powered document analysis, Gmail integration, web search, image understanding, and modular backend services.
- Designed scalable frontend and backend components with reusable architecture, environment-based configuration, exception handling, and persistent chat history.
- Deployed the application on Streamlit Community Cloud and managed the complete development lifecycle using Git, GitHub, and modern software engineering practices.

AI Research Assistant

- Developed an AI-powered research assistant using Python, LangChain, Google Gemini API, and REST APIs for automated report generation.
- Built a modular backend integrating web search, document generation, PDF export, file management, and reusable service components.
- Implemented secure API authentication, exception handling, input validation, and scalable application architecture following software engineering best practices.
- Generated structured research reports in PDF and text formats while managing the project using Git and GitHub.

Real-Time Pothole Detection Using Raspberry Pi and YOLOv7

- Developed a real-time pothole detection system using Python, OpenCV, YOLOv7, and Raspberry Pi.
- Integrated live camera input with optimized deep learning inference for edge devices.
- Improved detection performance through image processing, debugging, and performance optimization.

Obstacle Avoiding Robot Car

- Developed an autonomous robotic system using Arduino, embedded programming, and ultrasonic sensors.
- Implemented Bluetooth, voice-controlled navigation, and real-time obstacle avoidance algorithms.
- Integrated sensors, actuators, and motor drivers while performing testing, debugging, and hardware validation.

Wireless Mobile Charging

- Developed a wireless power transfer prototype using inductive coupling principles.
- Designed and tested transmitter and receiver circuits for efficient energy transfer.
- Optimized charging performance through circuit analysis and hardware testing.

EDUCATION

Teegala Krishna Reddy College of Engineering and Technology — **2022 – 2026**
Bachelor of Technology in Electronics and Communication Engineering — CGPA: 8.46

Gauthami Junior College — **2020 – 2022**
Intermediate (MPC) — Score : 92.4%

Sri Vani High School — **2019 – 2020**
Secondary School Certificate (SSC) — Score : 100%

CERTIFICATIONS

- **Salesforce Certified Agentforce Specialist**
- **Getting Started with Artificial Intelligence** – IBM SkillsBuild
- **Python Programming Bootcamp** – Udemy
- **Cyber Job Simulation** – Deloitte

ACHIEVEMENTS

- Founded **Neuronics**, the college robotics club.
- Received recognition for the **Best Mini Project** and **t**hroughout department.
- Contributed to the college magazine for **two** consecutive years.